

ISLAY PORT ELLEN, United Kingdom

WMO: 031050

Lat: 55.683N

Lon: 6.250W

Elev: 56

StdP: 14.67

Time Zone: 0.00 (EUW)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	28.1	30.8	23.1	17.4	30.8	25.4	19.5	33.7	59.8	44.4	51.4	45.5	5.9	50	0.781

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	68.5	61.2	65.5	59.7	63.4	58.5	62.6	66.6	60.9	64.0	59.6	62.1	13.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.8	79.7	64.4	59.4	75.8	62.4	58.3	72.9	61.0	28.2	66.8	27.1	64.2	26.2	62.4	72.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 42.6	34.6	29.9	DB	23.7	75.1	2.8	3.4	21.6	77.5	20.0	79.4	18.4	81.3	16.3	83.8	(4)
(5)			WB	22.6	66.2	2.8	2.4	20.6	67.9	18.9	69.2	17.3	70.6	15.3	72.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.9	42.1	41.6	43.1	46.4	50.3	54.5	57.1	57.0	54.8	50.7	45.9	43.1	(6)	
	DBStd	7.06	4.54	4.43	4.27	4.24	4.56	3.59	3.53	3.01	3.62	4.36	4.72	5.07	(7)	
	HDD50	1278	244	237	217	123	48	5	0	0	6	45	135	218	(8)	
	HDD65	5873	709	656	680	558	455	315	248	250	306	443	573	680	(9)	
	CDD50	880	0	1	2	14	58	140	219	216	150	66	11	3	(10)	
	CDD65	3	0	0	0	0	0	0	2	1	0	0	0	0	(11)	
	CDH74	10	0	0	0	0	2	1	6	1	0	0	0	0	(12)	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	14.5	17.7	16.0	15.8	14.9	14.1	11.9	10.9	11.8	13.4	15.3	16.1	16.3	(14)	
(15) Precipitation	PrecAvg	51.7	5.8	3.6	4.5	3.0	2.3	2.8	3.5	4.1	5.1	6.0	5.7	5.6	(15)	
	PrecMax	67.8	9.2	9.1	8.7	6.6	6.1	5.7	6.6	8.6	9.1	11.4	9.1	9.9	(16)	
	PrecMin	40.7	1.0	0.1	1.1	0.3	0.4	1.2	1.2	0.9	1.0	1.7	2.0	2.2	(17)	
	PrecStd	5.6	2.1	1.9	1.7	1.5	1.3	1.1	1.3	1.9	1.9	2.1	1.8	2.0	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.9	52.6	56.2	63.1	71.3	70.7	74.1	70.2	65.6	61.2	57.1	54.7	(19)	
		MCWB	50.9	50.1	50.1	53.0	59.4	61.8	65.4	63.7	61.1	57.6	55.2	52.7	(20)	
	2%	DB	51.1	50.8	52.9	58.6	65.0	66.4	68.6	65.7	63.2	59.1	55.4	53.1	(21)	
		MCWB	49.8	48.9	49.1	51.0	56.1	59.3	62.4	60.7	59.4	56.4	53.4	51.4	(22)	
	5%	DB	50.0	49.4	50.9	55.2	60.9	63.4	65.3	63.5	61.6	57.9	53.9	51.7	(23)	
		MCWB	48.9	47.8	47.9	49.7	53.8	57.5	60.3	59.3	58.3	55.4	52.1	50.2	(24)	
	10%	DB	48.9	48.1	49.4	52.9	57.7	61.0	62.9	61.9	60.2	56.7	52.8	50.3	(25)	
		MCWB	47.6	46.5	46.7	48.3	52.0	56.0	58.5	58.2	57.3	54.3	50.9	48.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.3	51.1	52.1	54.8	60.3	62.9	65.7	64.4	62.4	58.7	55.7	53.1	(27)	
		MCDB	52.0	51.7	53.7	58.8	68.3	69.3	72.2	69.1	64.0	59.9	56.5	54.0	(28)	
	2%	WB	50.2	49.4	50.1	52.3	57.5	60.3	63.0	61.9	60.4	57.3	53.9	51.8	(29)	
		MCDB	50.8	50.3	51.6	56.8	63.6	64.8	67.4	64.9	62.1	58.5	54.8	52.8	(30)	
	5%	WB	49.2	48.1	48.5	50.6	54.7	58.3	61.1	60.3	59.2	56.1	52.7	50.4	(31)	
		MCDB	49.9	49.2	50.2	54.1	58.9	62.2	64.4	62.5	60.7	57.4	53.8	51.5	(32)	
	10%	WB	47.8	46.8	47.3	49.2	52.9	56.7	59.3	59.1	58.1	54.9	51.1	48.9	(33)	
		MCDB	48.8	47.8	49.0	52.1	56.2	60.0	62.0	60.9	59.6	56.5	52.5	50.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.7	8.8	10.2	11.8	12.1	11.5	10.9	10.1	9.7	9.1	8.6	8.4	(35)	
		MCDBR	7.2	8.1	10.9	14.2	15.8	15.4	14.1	11.9	10.2	8.5	8.1	8.0	(36)	
	5% WB	MCWBR	7.3	7.3	8.4	9.0	9.6	9.8	9.2	8.1	7.7	7.2	8.1	8.1	(37)	
		MCWBR	6.9	7.5	9.1	12.1	13.6	13.4	12.5	10.3	9.2	7.8	8.1	7.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.345	0.353	0.372	0.384	0.388	0.380	0.382	0.369	0.366	0.350	0.335	0.332	(40)	
		taud	2.197	2.291	2.306	2.321	2.361	2.422	2.420	2.472	2.463	2.416	2.272	2.171	(41)	
	Ebn at Noon		170	223	250	265	271	275	271	268	252	224	175	143	(42)	
		Edh at Noon	19	25	31	35	36	34	34	30	27	22	17	16	(43)	
(44) All-Sky Solar Radiation	RadAvg		150	367	722	1210	1563	1579	1477	1193	807	438	193	108	(44)	
	RadStd		17	35	55	82	134	155	110	92	51	45	18	11	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (41 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page